Crib Sheet: Cirrus MPI Exercises for Online Course

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1 Cirrus account

This document assumes you have already been given access to a login account on Cirrus. In the instructions below, you should replace `username` with your own login, and the project `taXXX` with your own project (e.g. `ta012`).

2 Logging on

Use your username to access Cirrus: `ssh -XY username@login.cirrus.ac.uk`

You will be prompted for the passphrase for your SSH key and then for your MFA time-based one-time passcode (TOTP). Both are required.

You should then change directory to the `/work/` filesystem: `cd /work/taXXX/taXXX/username/`.

3 Obtaining source code

Remember that the compute nodes on Cirrus cannot see the `/home/` filesystem (your default home directory) so you should ensure all your files are in `/work/`.

Any source code will be available as tar files from the relevant course pages. Here we will use the `MPP-templates.tar` file as an example.

Download it to your local machine, then copy to Cirrus using

```
scp MPP-templates.tar username@login.cirrus.ac.uk:/work/taXXX/taXXX/username/
```

Now unpack the file: `tar -xvf MPP-templates.tar` and change directory into it: `cd MPP-templates`

4 Compiling code

The default compiler environment on Cirrus is `PrgEnv-cray`. To use the GNU compilers instead, load the GNU programming environment:

```
module load PrgEnv-gnu
```

You can compile the C, C++ and Fortran codes directly:

```
cc  -o hello hello.c
CC  -o hello hello.cc
ftn -o hello hello.f90
```

or using the supplied Makefiles

```
make -f Makefile_c
make -f Makefile_cc
make -f Makefile_f90
```

The compiler wrappers `cc`, `CC` and `ftn` are the recommended way to compile MPI code on Cirrus.

5 Running

To submit a job to the SLURM batch system: `sbatch cirrusmpi.job`

This submits to the short queue for rapid turnaround, and will run the program `hello` on 4 processes.

The batch system will respond with a unique ID for your job, e.g: `Submitted batch job 123456`

You can monitor the progress of your job with `squeue -u username`. The letters "PD" indicates the job is queued (i.e. pending), "R" that it is running, and no listing that it has finished.

When the job has finished, the output will appear in a file called `hello-123456.out`

You can kill a queued or running job using: `scancel 123456`

To change the number of MPI processes, run a different program or use a different queue then you will need to edit the SLURM script `cirrusmpi.job`.

6 Interactive Access

It is possible to submit an interactive job to SLURM. Your job will still have to wait for resources to become available, but screen output will appear in your terminal in real time rather than being redirected to a file.

```
srun --partition=standard --qos=short --reservation=shortqos \
--time=00:01:00 --unbuffered --cpu-bind=cores --nodes=1 --tasks=4 ./hello
```